Teaching Electro-Acoustic (EA) Composition to the Uninitiated

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Abstract

This paper looks at the processes used in teaching EA composition to students in the Production Centre at the Victorian College of the Arts. This situation provides a unique and interesting set of problems and opportunities, not least that many of the students have little to no prior interest in EA composition per se. This lack of interest gives the most exciting opportunities to a teacher, the two most pressing being to develop strategies to create an interest in EA and then developing a sense in each student that it is possible for them to be an effective composer in this field.

The VCA provides a rich environment for this to happen as people actively involved in the creative process and who are continuously producing works surround each student. This fever of creativity rubs off on students, who may initially come to learn the more prosaic aspects of theatre production and feel less able or inclined to approach their studies from a creative and personally expressive position. Fortunately this position changes during the student's tenure to that of being a vital cog in a creative engine or, more often, as an inspiration for and developer of fine artistic creations.

The topics discussed here include: the different processes used in introducing composition tools; developing an aural acuity and listening skills; introducing EA concepts and confidence with software; the collaborative approach to composition required by the situation; and how the general audience receives the EA offerings of the students. The software and hardware available, and the reasons for those choices, in the Production Centre is also discussed.

The creative milieu

As the Production Centre supports the Drama, Dance and Music schools, and to a lesser degree the Art and Film and Television schools, the opportunity and the expectation for students to create sound works to be used in each of these media is plentiful. Within this milieu each student is confronted with the varied sonic needs of each art form. Filling these needs can range from providing banal "sound effects" for a theatre piece to a complete EA composition for a dance piece, or vice-versa. This process is done in collaboration with directors, choreographers, and any other commissioning artist.

A major benefit for the EA composer when observing practitioners from other, non-musical, art forms is that the compositional and structural concerns and approaches unique to these practitioners are shown to the EA composer.

The students initial interest

When a student first enters the Production Centre at the VCA he or she may have minimal interest in the EA world, which is what prompted the title of this paper. The students main areas of work while at the VCA are in theatre and dance for which they gain expertise in all areas production; set design, lighting design and operation, costume making, stage and production management and sound design and operation, which is what this paper addresses.

In the structure of the course students spend first year developing skills in all of these areas, then select two that are of special interest to study in second year and then spend much of third year seconded to companies and acting in their chosen field. Their initial understanding of sound deliberately produced to create an effect may be that it is just that, a "sound effect" directed by the script or director. Music for a theatrical event may be what is heard as the patrons enter and leave the auditorium or music from a CD that is considered to enhance the action of the play at strategic points in the narrative. Often the main interest of the students when entering the course is to develop skills sufficient to ensure that they will be capable of working in theatre or similar areas when they have completed their course, and the Production Centre at the VCA has an excellent record in its graduates working in their chosen fields.

For this reason production students often begin from a physical point to their work. Questions such as "Do I need three phase power for my sound application?", "Can I rearrange the tracks on a Minidisk player?" and "What is the best reverb setting for this sound?" have been asked. The first two questions are to do with the physical needs of setting up a sound system for a performance production, they are questions do not require any personal or artistic input from the student. This approach to sound making can result in a prosaic and less adventurous, creative, or expressive approach to developing a sound design for a theatre or dance production. The third question presumes that sound can be affected and presented in a way as to change that sounds nature and the listener's perception of it; it is the beginning of EA composing. From this point the possibilities in manipulating the aural environment to create an affect in the listener become apparent and available to the student. Now the student begins their personal exploration and development as a composer.

The heuristic nature of each students development as an EA composer

In this situation students develop as composers mainly though experimentation. They have a few tools available to them which allow sound manipulation and fit two categories: real time manipulation, for use in the field; and computer based collation and manipulation, which allows the composer to offer a more finished product to the director, choreographer or whoever is commissioning the work. Students are encouraged to create their own sounds through recordings they do themselves and through manipulating the work of others, either commercially available sound effects or the music composed by others.

By making their own recordings skills in studio and field recording techniques are developed. The range of recording equipment available is of very high quality and, though limited in numbers, very broad in scope. Through the use of different microphones and their placement skills in critical listening and sound manipulation are developed. An example of this is the degree of manipulation that is by simply changing the pick-up pattern of a microphone from hyper-cardioid to omni-directional in the same space. Another example of microphone technique which, as a by product develops a more acute sense of listening, is through using shotgun microphones being trained on sources that are not usually heard by themselves, such as the sound of the device on a tram that connects it to the overhead electric wire. This introduction to the more unusual sounds around us creates a wider sonic pallet for the budding composer to use and inherently gives a sense that sound in itself is an easily manipulated source which can be used in creating an affect in the listener.

When students start manipulating the work of others they begin to develop compositional strategies. This is particularly interesting when working with pre-recorded compositions. In this case there is an already existing musical framework that has to be recognised, assessed and then subverted to the needs of the students work. Much of this recognition and assessment is done subconsciously and intuitively without the possible benefit, or obfuscation, of traditional musical thinking, the students simply try to achieve an appropriate aural environment for the commission. From here the students model personal compositional strategies from the pieces they are using, often with the sense that they have discovered the particular strategies themselves.

Composition from an aural position

Most students begin their development as a composer from a position that may best be described as "I may not know music (sound) but I know what I like. This can be a very powerful position for an EA composer to come from. As the broad palette of sound itself does not readily fit the concepts of counterpoint, harmonic rhythm, tension and resolution, musical form, set and specific pitch or any other traditional musical concepts or compositional devices as traditional instrumental music, these concepts or devices do not encumber the nascent EA composer. This does not mean that each of these devices are not used when composing the sound track for a dance or aural environment for a play, just that they are used with the aural affect on the dance or play being the prime consideration. The interesting phenomena is that students readily use these compositional devices from the point of having invented them in order to fulfil the necessities imposed by the commission they are working on, and they then assess the usefulness of each device within the context of that commission.

When teaching these devices are not ignored but introduced as examples present themselves and when problems in a specific commission arise that may best be solved with these devices. By immediately applying musical devices when they are required students generally gain a good grasp of the essential ideas and purposes behind each device and their possible applications. From this point they then develop expertise in areas of traditional musical composition and hone this expertise through listening to the results. Conversely, when the prime relationship to the composition is to serve and enhance another art form issues of compositional technique, style and genre carry less weight and may even not be best not considered.

There are also special problems and questions that may arise for the more experienced EA composer; problems as varied as the quality of recordings to be manipulated or having to learn the intricacies of a computer language, and questions such as: are sounds of similar envelop characteristics more closely related than sounds of similar timbre characteristics; is sound grouping, and its inherent problems, relevant to the current, unique commission and resulting composition; how important is the metaphoric qualities of a particular sound in the composition; and do the metaphoric qualities of a sound mask other qualities of that sound. These questions are tackled by the student within the context and confines of a well-defined commission. It is these confines that allow the developing EA composer to answer these questions first through analysing the work and its relevance to the commission themselves and then through the filter of collaboration with the director, choreographer or musician they are working with.

Non-musical media as the structural element

The sound design student and their collaborator develop EA compositions that usually have as their source a non-musical inspiration. This inspiration may be a choreographer's assignment, a sculptor's presentation, a score or Foley for a film, or a text and its interpretation into a play. Through working with artists in these various fields Production Centre students gain an insight into the many unique, and universal, problems and the possible solutions, encountered in the compositional processes of each of these art forms.

The composer David Ward-Steinman explored the structural aspects of various art forms and, at the time he was at La Trobe University in 1989, had found that there is a structural similarity between art forms, they generally follow a simple ABA structure. In its simplest form the observer is taken from a point of rest to a point of activity which then resolves in a return to a point of rest. This finding is becoming increasingly arguable in the wonderfully diverse time we live in, but can hold true if given considerable latitude in interpretation. The salient point for the EA composer is that each art form will have its own techniques of navigating this ABA, or any other, structure and the composer is then able to learn and apply these techniques to their own creative pursuits.

To use a film as an example of this process: The first step is to examine the intentions of a script and how its structural aspects reflect that intention. This could be the style of language used, the points at which characters are introduced and removed, the changes in the nature of the dialogue, or where the pivotal points in the script reside. The next step is to develop a sonic palette which reflects the intention and the final step is to compose this palette into a sound-scape which enhances both the scripts intention and its structure.

Most films will usually have a point close to the beginning where characters are introduced and the scene is set, followed by a period of exposition where each character is developed and the audience given a more detailed view of their motivations. This period of acclimatisation and rest is then followed by a disruption generated either by the characters themselves or by an unexpected outside agent. Finally the characters negotiate a method of returning to a state of rest at which point the play ends; Ward-Steinmans ABA structure.

This is an extremely simplified analysis, and of course the devil and delights of any artistic expression lie in the detail. The emerging EA composer can give detailed study to the script and, with its creators and interpreters, develop a coherent composition based on the details of the films structure. This relieves the composer of responsibility of developing an idea from scratch, attending to the minutiae of its development alone, its presentation and so on. With these activities accounted for the student can devote their attention to developing an understanding the needs of a piece and to explore their own approaches to fulfilling these needs.

Once a composition has been made the students are able to see the effect of their work, first on their co-participants and then on an audience proper. While the presentation of the composition is usually part of a multi-media event the importance of its contribution to the event is always evident and acknowledged. The VCA is an excellent forum for the emerging artist, it provides a very encouraging environment in which every attempt is made to enhance the talents of each student while at the same time not abiding charlatanism or half-heartedness.

Broadening the audience

The audience base for electronically manipulated sound and EA composition is developing through the regular use of this media in the performance arts. As creators of performance art become more aware of the possibilities available through electronic sound manipulation they become more prepared to both commission works from EA composers and are more able to communicate economically with composers. This is seen in the preparedness of choreographers at the VCA to leave the tradition of using off the shelf pre-recorded music to developing dance specific EA compositions for their work. Performance artists will also try their hand at composing themselves but when doing this most realise the benefits of working collaboratively with a composer who can introduce musical strategies and who can often also double as a technician.

This collaborative approach to EA composition gives the student a unique opportunity to educate, or at least inform, directors, choreographers and traditional musicians and composers, in the possibilities available within the world of EA. Students and their collaborators initial explorations often revolve around the simplest alterations to natural sounds, such as pitch shifting or imposing spatial motion on the sounds. This simple exploration creates a sudden insight into the impact of sound and the possible effect it can have.

Through this introduction to EA practitioners of other performance arts become more versed in the EA arts, often searching out EA music to use in their performances and composers to commission. The result is an increasingly wider audience than the usual enthusiasts and, what is more important, a non-musical audience.

Audio technology in the Production Centre

One of the most pressing requirements of the sound students at the VCA Production Centre is that works must be completed quickly. There is little time to develop software or technology, this is not in the brief of the Production Centre, nor is there time to use software or technology that requires either a long time to learn or produce sounds. For this reason two standard audio editing systems, Cool Edit on a PC, and ProTools on a Macintosh are used. These two platforms were chosen because introduce the students to industry standards and because they offer fairly generic interfaces and procedures for assembling sounds.

The sound manipulation software currently used in the Production Centre includes the standard set of plug-ins bundled with Cool Edit Pro. When used effectively and imaginatively, these algorithms can produce many varied and interesting sounds by altering any kind of pre-recorded sound wave file. On the Macintosh SoundHack, which have different attributes to the Windows plug-ins available in Cool Edit, is used as the sound manipulation tool. This programme requires more time to explore, operating slower than real time, and a greater understanding of the EA lexicon. When this understanding is developed it proves to be a more powerful sound manipulation programme and offers scope for more conceptual and well developed composition processes.

Other programmes include AudioMulch, which is very powerful programme that allows quite sophisticated signal modification, it can also be used to quickly generate dance club style music. This attribute is quite exciting to the early EA composer as it can provide a composition of their own that often fits with their personal aesthetic. This instant gratification helps

develop confidence to explore the programme further and to develop skill in using the programme.

A visual composing programme called Coagula is also extensively used by students. This programme allows a student to draw an image on the computer screen that it then translates into an audio signal through sine wave synthesis. Coagula has wonderful advantages in that it too offers instant gratification as well as drawing a direct relationship between a physical and a sonic gesture. This is particularly powerful when working with choreographers and other visual artists.

The hardware available includes a standard set of signal processors, digital and analogue mixing desks and sampling hardware. These items are very important to the developing EA composer as they are simple, often logical in their operation and allow the composer to improvise with sound. This process is specially useful when working on student drama productions as these often require the Production Centre student to improvise a sound track using CDs. The student is encouraged to manipulate the music from the CDs by cross fades, multiple sound tracks and signal modification. The degree to which a student practises and executes these manipulations is dependant mostly on their imagination. It has been noticed that students may first assume that the audio manipulation power available from a mixing desk and two CD players is minimal but after practice and exploration discover that they have a powerful audio manipulation system available to them.

This set of EA devices may seem quite limited at first glimpse but it does fulfil the requirements of a mostly real time studio for the EA composer. The very limitations of the equipment available require a reliance on imagination and artistry to generate high quality product and not a reliance on expensive hard or soft ware to produce interesting sounds. Other considerations when acquiring the equipment available has been chosen to give students as broad a range of the technology available in the performance art industry and that each student be able to equip themselves with a similar studio without too much expense and, as software becomes available it is included in the studios if appropriate.

Teaching strategies

As can be seen from much of the above it is the environment of the VCA and the unique place the Production Centre has in that environment that provides the greatest role in developing the emerging EA composer. Much of the teaching is done on an individual level, assisting students in the needs of each commission as it arrives. There is also a great deal of interaction and communication between the students regarding the discoveries they have made and the strategies they have developed. These strategies may be to do with compositional problems, developing ideas from and for commissions, exploring new ways to use software and to develop communication skills with other performance artists. This interaction between students is encouraged as a philosophy of the Production Centre and is a formal part of the teaching process.

When teaching sound design per se students are first given scripts as the starting point for an EA composition. They are encouraged to look at the script without consideration for any aspect of the performance other than sound. This gives each student an opportunity to look at the intentions and structures of the script and to let their imagination run free in enhancing those aspects.

The results are often astounding, basing the process on ideas and then discovering how to implement them encourages the student

to approach learning software and hardware in a very directed way. This means that little time is wasted in less useful learning and students are vitally interested in the information they receive.

Students are given very few assignments as most of their work is in the production of performances. The few assignments that are handed out are based on developing rudimentary ability with the technology and are assessed on the students musical use of the technology. This approach enhances the way students approach their commissions as they develop a more detailed awareness of the musicality of sound and a sense of musicality in the sound designs they make.

Most assessment in the Production Centre is based on how well the commission each student is given is fulfilled and this is assessed by staff in all areas of theatre production practise. The broadness in aesthetics of each staff member means that students have a variety of opinions from which to develop a greater sense of their work.

Conclusion

This approach to teaching EA composition does not follow the usual steps. Students are not considered to have any musical background or knowledge to be expressed through EA compositions, they are not expected to have any interest or ability in creating software for musical application, and these more typical aspects of EA teaching are not a high priority in the teaching process.

What is considered most important in the teaching is a sense that EA hardware and software are expressive tools to be used for artistic expression. From this base many students are developing an interest in the more traditional areas of EA simply for the purpose of increasing their creative abilities.